

Ali Rezaee

📍 Mashhad-Iran ✉ a.rezaee94@gmail.com ☎ +98-915-888-0658 in ali-rezaee-ferdowsi 🌐 AliRezaee86

Summary

Water resources management specialist (**M.Sc. Ferdowsi University of Mashhad**) with expertise in hydrological/climate modeling, machine learning, and big data analysis using Python/R. Current research focuses on extreme weather events (droughts, floods, heatwaves) and compound extreme events under climate change.

Education

- M.Sc. Ferdowsi University of Mashhad**, Water Resources Management Sept 2019 – May 2024
- GPA: 17.8/20
 - **Coursework:** Stochastic hydrology, Urban hydrology, Hydrological and climate models
 - **Thesis title:** Time series analysis of precipitation, temperature, and evapotranspiration in different regions of Iran from 1961 to 2020.
- B.Sc. Shahrood University of Technology**, Water Engineering Sept 2014 – May 2019
- GPA: 16.2/20
 - **Coursework:** Mathematics, Applied statistics, Hydrology, Groundwater, Hydro-climatology, Runoff modeling, GIS and HEC-RAS, Matlab programming

Skills

Python: Machine learning, Big data analysis (Pandas, NumPy, Geemap, SciPy, Scikit-learn, TensorFlow)

R: Statistical analysis, Climate data visualization

JavaScript: Geospatial scripting (Google Earth Engine)

MATLAB: Time series analysis, Wavelet transforms, Hydrological modeling

Geospatial tools: GIS, GIS-Pro, QGIS, CDO (Climate data operator)

Specialized software: GMS, SWMM, HEC-HMS, HEC-RAS, ENVI, GIT, Prezi

Communication & collaboration: Teamwork & leadership, Independent research & problem-solving, Scientific writing, Time management, Adaptability in fast-paced environments, Teaching and explaining complex technical/conceptual topics

Linguistic: English (Intermediate, currently participating in IELTS preparation classes), Kurmanji(Native), Farsi(Native)

Experience

- Ferdowsi University of Mashhad**, Teaching assistant Iran, Mashhad
June 2020 – Aug 2023
- Courses: Applied python programming in water engineering, RS & GIS, Stochastic hydrology
- Khorasan Regional Water Company**, Associate researcher Iran, Mashhad
Oct 2024 – Present
- Creating a groundwater database and modeling/predicting water levels for the province.
 - Investigating subsidence and proposing prevention strategies.
- The Water and Environment Research Institute at Ferdowsi University**, Associate researcher Iran, Mashhad
Oct 2024 – Present
- Collecting, validating, and preprocessing data of observational, remote sensing, and climatic models
 - Developing predictive and simulation models for water and environmental research
 - Conducting statistical analysis and integrating data for climate research insights

Publications

- * Using wavelet transform to analyze the dynamics of climatic variables; to assess the status of available water resources in Iran (1961-2020)** Aug 2024
Ali Rezaee, Abolfazl Mosaedi, Aliasghar Beheshti, Azar Zarrin
[Earth Science Informatics](#)
- * A copula-based framework for assessing heat wave extremes and probability of occurrence in Iran; implications for climate change** Apr 2025
Ali Rezaee, Mohammadreza Goodarzi, Hossein Etemadfard, Mohammad Alavizadeh
[Stochastic Environmental Research and Risk Assessment](#)
- * Machine learning insights into land surface temperature variability and prediction: A spatiotemporal approach with feature importance and uncertainty analysis** Feb 2026
Ali Rezaee, Mohammadreza Goodarzi, Mohammad Alavizadeh, Mojtaba Goldani
[Environmental Monitoring and Assessment](#)
- * Multi-timescale streamflow projections in human-regulated basins of southwest Iran using CWatM and explainable machine learning** May 2026
Mahdi Mohammadnezhad, Ali Rezaee, Mikhail Smilovic
[Journal of Hydrology: Regional Studies](#)
- * A Framework for Non-Stationary IDF Curve Estimation in Data-Scarce Regions: Evidence of Shifting Extremes in an Arid City** May 2026
Sara Attaran, Ali Rezaee, Mohammad Reza Nikoo, Badar Al Jahwari
Article in the Journal of Arid Environments is under-review

Projects

- Aquifer modeling of Najaf Abad plain, Isfahan** 2024
- Built and calibrated a MODFLOW-based aquifer model integrated with GIS to simulate groundwater flow and optimize model parameters based on pumping and recharge data.
- Urban heat island (UHI) analysis in Mashhad, Iran** 2024
- Implemented a satellite-based UHI assessment using MODIS data, GEE, and Python to investigate the relationship between extreme heat events, electricity demand, and grid reliability.
- Downscaling of ERA5 snow depth data** 2025
- Developed a machine learning framework to enhance ERA5 snow depth estimates by integrating remote sensing, topographic, and ground observation data for improved predictive performance.
- Land subsidence simulation framework in Mashhad aquifer, Iran** Sep 2024 - present
- Developed a hybrid ML-GWR model to analyze drivers of land subsidence.
 - Predicted future subsidence under groundwater level decline scenarios.
- Data Governance Framework for Water Resources and Environmental Management** Dec 2025 - present
- Reviewed global data governance practices and developed a five-dimensional framework with regional objectives.

Research Interests

- Extreme event modeling (Hydro-climatic) and Compound extreme events.
- Precise uncertainty analysis.
- Machine learning and deep learning applications in hydrology and climate science
- Remote sensing data and big data analysis